Vi-UNet and SegNet for Landslide Segmentation: A Dual-Model Approach to Hazard Mapping

Valikala Sudesh Chandra[1], Kanchumarthi Venkata Sudhanva[2], Badagala Naveen Kumar[3]
1. Computer Science and Engineering – AIML, VIT-A.P., sudesh.22bce7267@vitapstudent.ac.in
2. Computer Science and Engineering – AIML, VIT-A.P., sudhanva.22bce7409@vitapstudent.ac.in
3. Computer Science and Engineering – AIML, VIT-A.P., naveen.22bce7726@vitapstudent.ac.in

Abstract - Landslides are a major threat to safety and infrastructure in the disaster-prone region, hence requiring effective detection mechanisms. This research explores advanced deep learning models to be used for detecting landslides based on the usage of custom and pre-trained architectures. We derive and test two custom models: Vision-Transformer-based UNet (Vi-UNet) and SegNet, as well as pretrained models UNet and UNet++ comparatively. The dataset used in this study includes a curated dataset of annotated landslide imagery for enhancement, which is further preprocessed to improve data quality and hence model performance. Evaluation focuses on accuracy as the primary metric by which to compare the relative merits of these models in detecting landslide-affected areas. Experimental results clearly reflect that the proposed Vi-UNet is characterized by superior accuracy with its ability to capture fine-grained detail and global contextual information.
This work shows the promise of transformer-based architectures for geospatial analysis and forms a foundation for building efficient, high-accuracy models for natural disaster management.

Keywords: Landslide Detection, Vi-UNet, SegNet, UNet, UNet++, Deep Learning, Accuracy Metric, Vision Transformer, Semantic Segmentation, Natural Disaster Management, Geospatial Analysis, Custom Neural Architectures, Pretrained Models, Disaster Prediction, Image Segmentation

1. Introduction

Landslides can be considered one of the most significant natural hazards in mountainous and hilly areas, leading to the loss of life, damage to properties, and disruption of infrastructure. Landslide detection in real time is a key to successful disaster management and mitigation action. Traditional landslide detection depends on manual inspection of satellite imagery or aerial surveys, which are exhaustive in nature, consuming much time and being prone to human errors. Application of deep learning techniques in landslide detection has shown a lot of promise in recent times to automatically analyze large geospatial data sets very efficiently and at a scalable scale.
Among all the architectures of deep learning, CNN has a tremendous success rate when applied to image segmentation. These were originally designed for biomedical purposes. While UNet is being used for medical imagery, it has found great use in the application of remote sensing with landslide detection. The encoder-decoder network structure with skip connections, which preserve spatial information, makes it preferable for pixel-level classification tasks. Furthermore, the advanced versions of UNet, which are UNet++ and SegNet, go deeper and introduce additional skip pathways, which helps the model learn sophisticated features in input images.
However, traditional CNN-based models fail to incorporate long-range dependencies and global context in landslide detection tasks accurately, especially because the topographies and the different soil erosion levels across regions can be very complex. In order to respond to this, recent developments in transformer-based models, such as the Vision Transformer (ViT), were able to model the global context using self-attention mechanisms. Integrating ViT with CNN-based architectures would potentially improve segmentation accuracy by making the best use of both local feature extraction and global context modeling.
Here, we propose integrating Vision-Transformer-based UNet, (Vi-UNet) and SegNet as custom models for landslide detection. We then compare their performance with that of the widely used pretrained models, UNet and UNet++. Our objective is to evaluate the impact of transformer-based models on segmentation accuracy, using accuracy as the sole evaluation metric. This study aims to provide insights into the effectiveness of Vi-UNet and SegNet in detecting landslides and to contribute to the growing body of work on leveraging deep learning for geospatial disaster detection.
Related work in the area of landslide detection and deep learning-based segmentation models are reviewed in Section 2. Section 3 explains the method including model architectures, preprocessing data steps, and evaluation approach used. Results of experiments are presented in Section 4, and conclusions are provided in Section 5 and References in section 6

2. Literature Work

Figure 1. Landslides and Non-LandSlides

This problem of precise forecasting and landslide detection has led researchers to modify ML and DL techniques, by using various data sources and models in enhancing the accuracy of the predictions and relevance across regions.[2].Author Acharya explores landslide probability prediction through a wide range of machine learning algorithms such as Random Forest, XGBoost, K-Nearest Neighbors (KNN), Support Vector Regression (SVR), and Linear Regression. This technique utilizes various satellite data sources that employ feature selection techniques, such as Correlation-based and Mutual Information-based selection, to improve the model. Among the models, Random Forest has the highest performance with an MSE of 0.0011, and KNN also performed well combined with correlation-based feature selection which resulted in reaching MSE of 0.00139. Adding multi-sensor data along with soil moisture, elevation, slope, and precipitation to the models reinforces the predictive power from comprehensive environmental insights. One major limitation of the study is that for the model, it is largely sensitive to imbalanced datasets, not to mention its high dependence on high-quality data. The factors which limit such applications are very high demand for computation while handling large multi-source data and limited region-specific generalization towards a broader applicability of this model. Problems like outliers and noise in the satellite data reduce the reliability of the early warning that is provided by the model, thus demanding an immediate improvement in preprocessing techniques for data and optimization of models to be applied in real time.[8]. The authors of this article rely on space-area resources put together with usage of Geographic Information Systems (GIS) and Artificial Neural Networks (ANNs) for landslide risk determination through geospatial data sources, which include DEMs slope maps, and remote sensing data. The class imbalance concern is addressed using SMOTE Synthetic Minority Over-sampling Technique reinforcement of the model. This approach employs ANN and achieves an accuracy level of 75%, precision of 80%, recall of 70%, and an F1-score of 75%. Geospatial data improves the capability of the model to carry out the analysis of relevant critical environmental factors for landslide risk assessment. However, there are certain disadvantages of the research from ANNs that generally function as black-box models. This creates a problem of result interpretation and confines the information concerning causal factors of the landslide. Moreover, although it is a good balancer of classes using SMOTE, it even introduces the risk of overfitting and leads to probable inaccuracies. Required computational resources in the form of a strenuous workload and difficulties in achieving reliable real-time detection call for more sophisticated models. In this paper, four CNN-based deep models- U-Net and three more related models: LinkNet, PSPNet, and FPN-have been compared with ResNet50 as the backbone for landslide detection. Based on the conclusion made by the authors, the highest accuracy in the detailed image analysis through semantic segmentation is delivered by LinkNet with 97.49%, resulting in an F1-score of 85.7%. Experiments were carried out on hyperparameter tuning during the optimization process with an encoder ResNet50, which highly boosts the features' extraction ability thus improving the accuracy detection rate. Practically, insights involving landslide detection by satellite images using the Bijie landslide dataset are used. Improvements in real-time performance entail corresponding increases in computational costs, while the variability of spatial resolutions impacts the generalisability of models. The fine-tuning stage involves fewer images with less perfect boundary representation, mostly containing detailed information in detail. This further limitation makes CNN-based models inappropriate for use with real-time landslide detection over a large scale due to the need to have high-resolution and quality images.[4].Yang explores DL applications in landslide mapping by reviewing 77 DL-based detection methods that integrate high-resolution satellite imagery, CNNs, and U-Net architectures in features extraction and generalization. This method is hugely dependent on the multitemporal and multisensor remote sensing, like TripleSat and Landslide4Sense, to be able to enhance the overall accuracy of results in detection. Conversely, it is indeed true that CNN-based approaches using high-resolution images are associated with better performance metrics in terms of accuracy, precision, recall, and F1-score. Limitations of availability of data, inventory constraints, and the sensitivity to resolution reduce generalisability to larger regions.[11].Youssef presents a receptive interpretable Superposable Neural Network (SNN) that is conceived to enhance the optimization of landslide susceptibility by using feature-specific ANNs, along with the outcome of their combination. In such a framework, there is the clear identification of landslide's significant predictors such as slope-precipitation coupling. Results: SNN is validated using landslide inventories from the eastern Himalayas; it performs very well with excellent AUROC, accuracy, sensitivity, and specificity values that compare to traditional models but with much less complexity. Its limitations are the resolution data is many field measurements and reducing cross-regional generalizability. 6. It was proposed in a study by Ghorbanzadeh et al that deep learning and Object-Based Image Analysis can be integrated together for landslide detection, especially in order to reduce false positives in complex topography. The authors introduce a high variant of ResU-Net: an improved variant of the well-known U-Net architecture with residual blocks taken in order to progress more efficiently at training with capturing both low-level and high-level spatial representations in distinguishing landslide features from riverbeds, and features similar to terrains. In OBIA, meaningful objects in the context of spectral, textural, and contextual features are collectively aggregated by grouping pixels together in order to enhance the result of ResU-Net. In this context, the accuracy of the detection enhances if knowledge-based rule sets are applied over elements that include length-to-width ratios for filtering out positive false signals. With this, it was discovered that ResU-Net, OBIA, and the hybrid ResU-Net-OBIA compared results that identified the best accuracy was achieved using the method that incorporated two, providing an effective hybrid approach towards landslide identification 5. Zhang and Wang introduce a comprehensive review on the uses of deep learning networks in landslide detection and prediction, giving way to five broad categories: Classification, Detection, Segmentation, Sequence, and Hybrid-the ones appropriate for both remote sensing and geo-environmental datasets. The Framework utilizes supervised learning to classify regions into landslides and non-landslides, and the Detection Framework centers landslide localization in imagery through object detection models such as Faster R-CNN and YOLO. Segmentation models such as U-Net and FCN are also proposed for pixel-wise classification for detailed boundary mapping of landslide regions. Sequence frameworks, such as RNN and LSTMs, would deal with the time series data to trace the evolution of how the changes in the terrain have been in time, whereas hybrid frameworks are an amalgam of different approaches for more complex landslide predicting tasks. This paper also tries to provide the benefits of merging multispectral satellite and SAR data with geological and climatic conditions; thus, the conclusion here is that hybrid frameworks are promising but depend mainly on the availability of computational resources as well as good-quality labelled data to be really efficient. [3] .Azarafza et al. presented a hybrid model called CNN-DNN for landslide susceptibility mapping by comparing its performance with six models of traditional ML models, SVM, LR, DT, etc. Within this paper, spatial datasets are automatically rm-extracted with features and then classified by DNN using landslide susceptibility. Historical landslide data along with geomorphological, geological, and environmental factors along with anthropogenic influences such as proximity to roads and cities are used for training and testing the models. AUC, IR, and MSE are reflected to have shown the representation of CNN-DNN to perform drastically better than traditional methods with AUC reaching up to 90.9%. Although the study boasts good performance, limitations such as high computational cost and the challenge of interpretation in the "black-box" nature of the CNN-DNN show up. It is understood that with larger and higher-resolution data, CNN-DNN can even improve landslide predictions and applicability across different terrains.

3. Methodology

This methodology combines two advanced architectures, namely, SegNet and Vi-UNet, to derive their strength and create efficient models with high precision in pixel-wise classification. These models were specifically selected to overcome the challenges owing to complicated, high-resolution imagery so that relevant features in images could be efficiently segmented.

3.1 Data Collection and Preprocessing

The dataset, for this particular study, includes three main categories: images of landslides, their corresponding masks, and images of non-landslides. The images of landslides include masks for segmentation, which set each pixel as a landslide or background, thereby enabling precise pixel-level classification. Non-landslide images carry the same kind of imagery but with all the features of land slides absent to offer a contrast class for the model to distinguish between these two classes.

These images were found from Kaggle, a platform highly recognized for superior datasets within the machine learning community. Using the dataset from Kaggle ensures the reliability and diversity of the images, as they come from real-world scenarios, making them especially well-suited for training deep learning models in geographic and environmental applications.

High-resolution pre-processing was carried out on several key steps to ensure the data fit well into the input of the neural network. Data augmentation techniques including random rotation, horizontal and vertical flips, scaling, and color adjustment were applied to artificially increase the dataset and make the model more robust to variations in input. All input images were also resized to a uniform resolution of (224×224×3), and pixel values were normalized to the range [0, 1], ensuring consistency during training. This preprocessing pipeline helped in preparing a diverse and well-balanced dataset, mitigating the risk of overfitting and promoting generalization across different image conditions

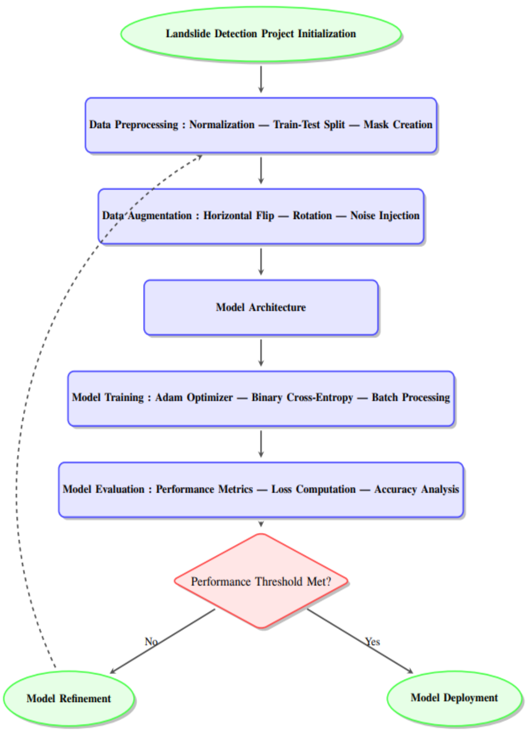

Figure 2. Workflow for LandSlide Detection

3.2 Model Architecture

The proposed approach integrates two powerful architectures: SegNet and Vi-UNet, both of which are tailored for semantic segmentation tasks but differ significantly in their underlying mechanisms.

3.2.1 SegNet Architecture: SegNet is a widely established architecture for semantic segmentation, suitable for applications that involve fine-grained pixel-level classification. The SegNet model adopts the encoder-decoder structure. It accepts input images and generates segmentation maps fast by processing the input image with an encoder comprised of multiple convolutional layers sequentially extracting abstract features from the input image. These convolutional layers are followed by max-pooling operations downsample the feature maps thus enabling the model to learn hierarchical representations of the data. For the sake of the SegNet, this encoder is set up to begin with 64 filters in the first convolutional block and gradually increase to 128 filters for deeper layers; such an approach would allow the model to capture detailed spatial features while keeping its computations efficient.

The decoder recovers the resolution in the feature maps. For that, it uses layers of upsampling followed by convolutional layers that refine the feature maps after the upsampling layers.The output layer is a single 1×1 convolutional layer with a sigmoid activation, producing a binary segmentation mask for each pixel.

SegNet's architecture was chosen for its efficiency in encoding and decoding image features, making it a reliable baseline for semantic segmentation tasks. It is particularly effective when dealing with high-resolution images, as it maintains spatial accuracy without excessive computational overhead.

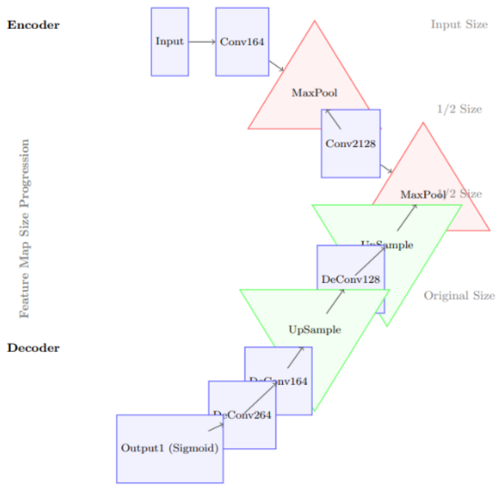

Figure 3. SegNet Model Architecture

3.2.2 Vi-UNet Architecture. The model combines a hybrid Vision Transformers (ViT) architecture with a U-Net, an architecture used to capture both local and global spatial dependencies. The encoder part of the Vision Transformer splits the input image into fixed-size patches. These patches are linearly embedded and then passed through several transformer layers. Traditional convolutional models have been focused on working with pixels in local relationships, while transformers are fantastic at modeling long-range dependencies between pixels that are distant. This means that a transformer can infer broader contextual information. This helps significantly in segmentation tasks, especially in complex environments where understanding the relationships of distant features is required for proper segmentation.

The ViT encoder is combined with a U-Net decoder, intended for progressive upsampling of the feature maps and refining the segmentation mask. The decoder applies a number of transposed convolutions to recover the resolution of the image while boosting the details in the segmentation output. Skip connections between encoder and decoder layers are ensured so that fine-grained spatial information is preserved throughout the upsampling process. At the final stage, a segmentation mask is produced as the output of 1x1 convolution followed by sigmoid activation.

Vi-UNet was chosen in order to combine the global contextual information provided by the Vision Transformer with precise, localized feature restoration as done by the U-Net decoder. This combination allows the model to capture fine-grained details needed to perform several state-of-the-art localization tasks and broader contextual relationships.

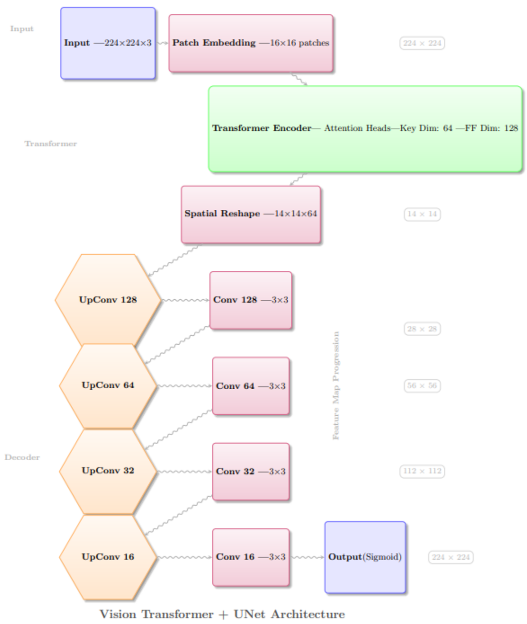

Figure 5. Vi-Unet Model Architecture

3.3 Model Training

Both the models, namely SegNet and Vi-UNet, have been trained using the TensorFlow/Keras deep learning framework by following a generic training procedure. The most important part of the training pipeline adopted image data generators, which allowed for feeding augmented images into the model efficiently. These generators provided real-time augmentation of the data to enable the model to expose variations in input conditions, thus generally improving its ability to generalize.

The training phase relied on the compiler function of the Adam optimizer, using an initial learning rate of 0.001 and altering it according to the model's performance. The loss function selected for both architectures was binary cross-entropy, since the task is a binary segmentation: foreground vs. background. This loss function penalizes incorrect classifications of pixels so the model can correct and get progressively better at the task at hand. The training process was run over 20 epochs with early stopping to prevent overfitting and to ensure improvement in performance on the validation set.

The batch size for training was set to 32; this is a good choice, as it still balances the need to process enough data per iteration against the constraints of limited memory. The abilities of the models are rated by how accurately they classify pixels. The main performance metrics here was accuracy.

3.4 Model Evaluation

After training the SegNet and Vi-UNet models, an evaluation was done on an independent validation set to test their segmentation performance. These were images accompanied by their respective segmentation masks which the models had not seen at training time. Accuracy was mainly the performance indicator for both models, since it represents a direct measure of percentage of correctly classified pixels in the segmentation maps. Accuracy is the pixel-wise classification evaluation metric that has been defined as the ratio of correctly classified pixels, including landslide and non-landslide, to the total number of pixels in the image. As the segmentation task was binary in the context of landslide vs. non-landslide, accuracy was computed by comparing the predicted masks with the ground truth masks.

Accuracy=True Positives (TP)+True Negatives (TN)/Total Pixels

Where:

True Positives (TP): Correctly classified landslide pixels, where the model correctly classifies pixels as land-slides.

True Negatives (TN): Correctly classified non-landslide pixels, such as background pixels (where pixels are correctly classified as background).

Total Pixels: The total number of pixels in an image.

4. Results

Metric	SegNet	U-Net	Vi-U-Net
Training Accuracy	93.36% to 94.80%	Improved, final at 77.59%	93% to 94%
Training Loss	0.2904 to 0.1238	Final loss at 0.4732	0.2904 to 0.1420
Validation Accuracy	100% (drops to 95.50% by Epoch 20)	Reached ~77.59% by Epoch 20	100% (drops to 95.50% by Epoch 20)
Validation Loss	Fluctuates, final at 0.1286	Final loss at 0.6878	Fluctuates, final at 0.1286

Table 1. Summary of custom Models and pretrained Models

In this study, three deep learning models-SegNet, U-Net, and Vi-U-Net-were tested with a landslide detection task. The models were trained and tested over 20 epochs, and their performance was analyzed using accuracy, loss, F1 score, precision, recall, and time per step.

SegNet showed steady progress of both training and validation metrics. Training accuracy began at 93.36% and progressed to 94.80% in the last epoch, where loss in training decreased from 0.2904 to 0.1238. Validation accuracy reached 100% in most epochs but slightly fell to 95.50% by the final epoch. The validation loss varied throughout but peaked in the final epoch at 0.1286. The time per step was steady and lied between 89 ms to 121 ms.

U-Net showed a good improvement in training and validation performance. The training accuracy was raised to 77.59%, with the final training loss showing as 0.4732 and the validation loss being 0.6878. The model's F1 score and precision also improved with the training F1 score improving to 0.7998 and the validation precision improving to 0.8656. The recall remained very high throughout the training process, being nearly 0.9 in all epochs. However, U-Net's final accuracy was lower compared to SegNet and Vi-U-Net, indicating the need for further optimization.

Training accuracy of Vi-U-Net was similar to that of SegNet with its increase over the epochs from 93% to 94%. By the final epoch, the validation accuracy dropped slightly by 95.50% as the training loss went down from 0.2904 to 0.1420. The loss in validation remained low but fluctuated between 0.0214 and 0.1286. Overall, Vi-U-Net showed stable performance with little fluctuations in the validation accuracy and loss towards the final epochs.

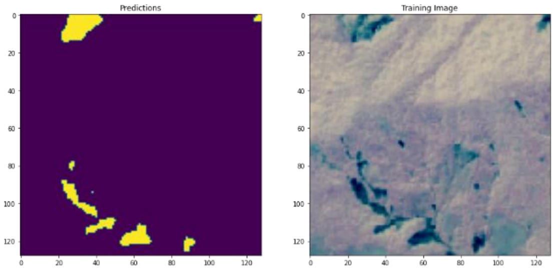

Figure 6.1

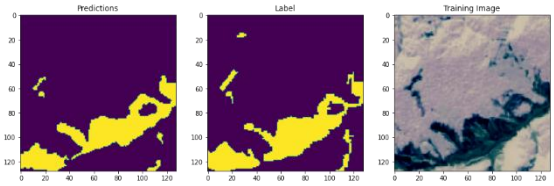

Figure 6.2

Figure 6.1, 6.2 depicts the Masking of Satellite images

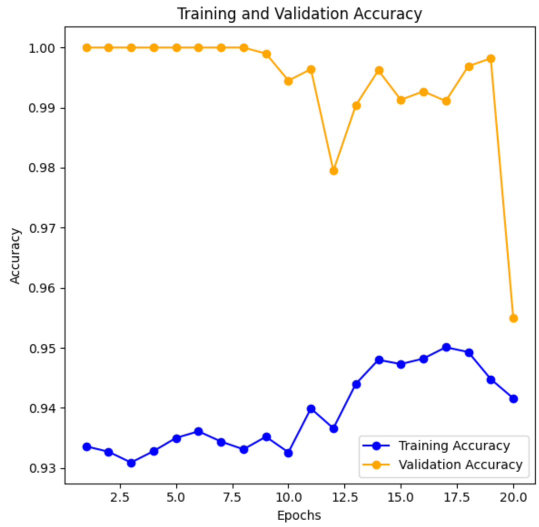

Figure 7.1

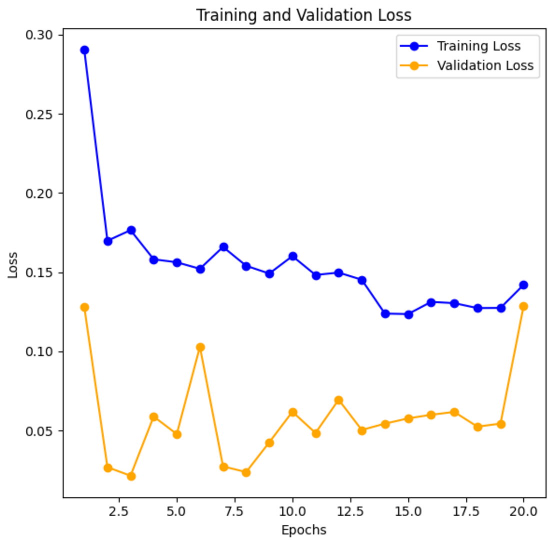

Figure 7.2

Figure 7.1, 7.2 depicts Vi-Net Performance

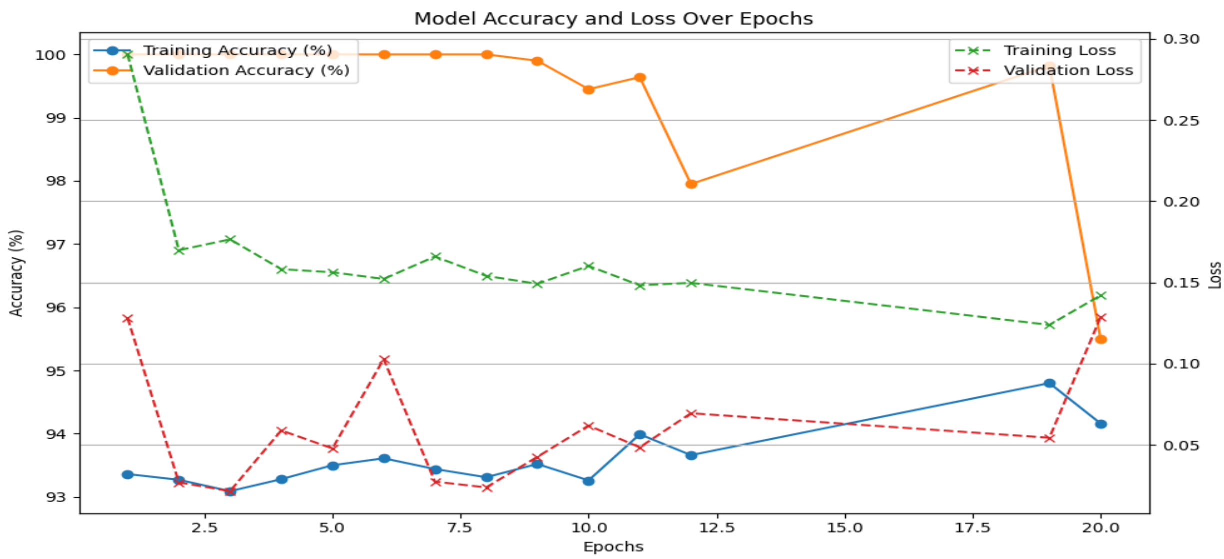

Figure 8. SegNet Performance

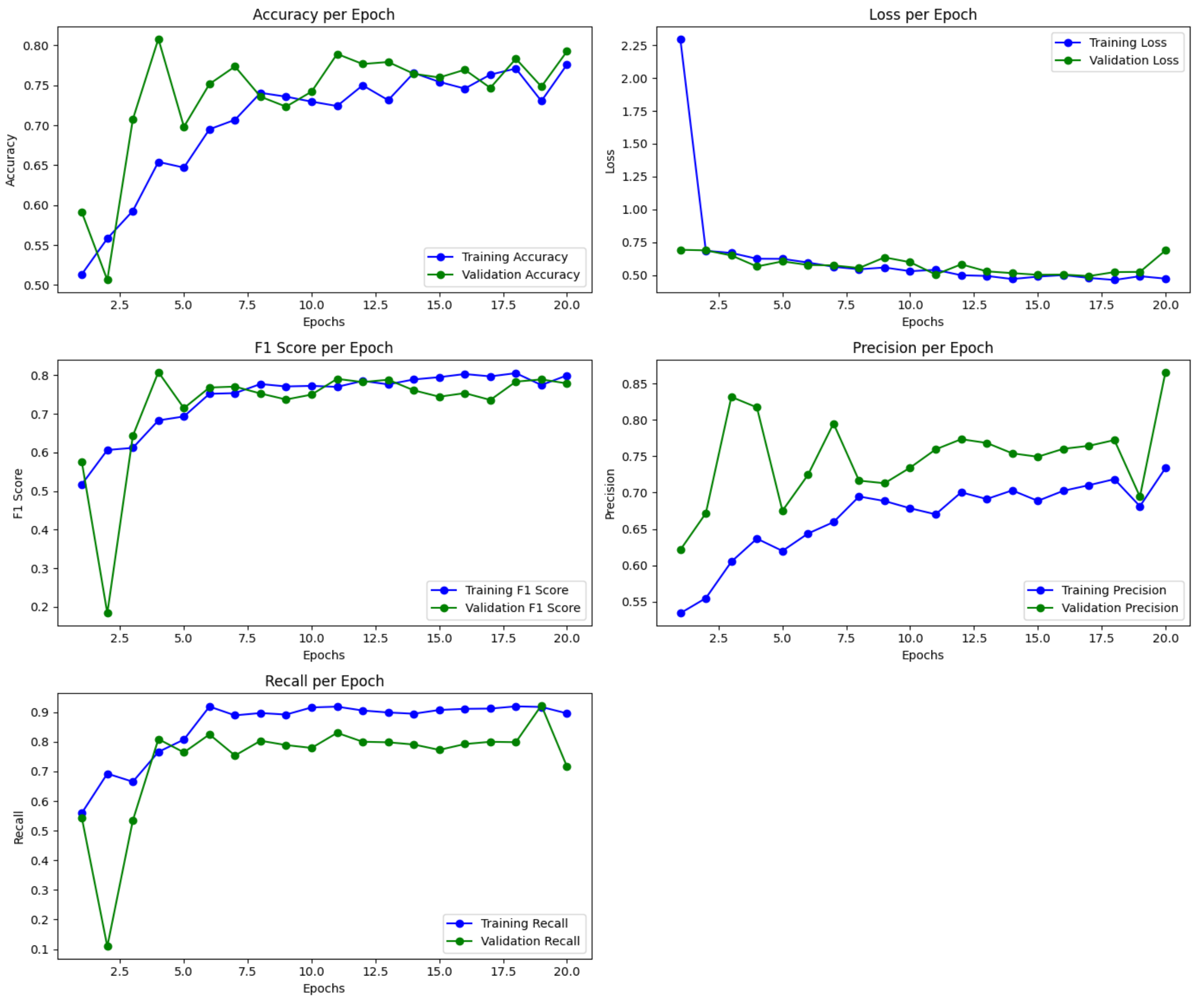

Figure 9. Unet Performance with regards to different Metrics

5. Conclusion

This research has brought a comprehensive evaluation of three deep learning architectures, namely SegNet, U-Net, and Vi-U-Net, applied to the critical task of landslide detection. This was performed through rigorous training and testing for over 20 epochs on several metrics including accuracy, and loss, valuable insights into the suitability of the models for geospatial analysis and disaster management applications.

The results have shown that both SegNet and Vi-U-Net actually outperform U-Net in the accuracy and stability perspectives. In fact, the highest final accuracy obtained was from SegNet with a value of 94.80%. Vi-U-Net followed closely with 94.00%. Both models have demonstrated strong training and validation performance with minimal fluctuations in both accuracy and loss. Interestingly enough, SegNet performed the best in terms of overall accuracy, having a consistent improvement curve both during training and the validation phase; hence, it was the most reliable for landslide detection tasks in the study. Though Vi-U-Net lagged slightly behind SegNet in terms of final accuracy, it had superior stability in validation loss, making it particularly valuable in real-time, high-precision applications, where consistency counts.

In contrast, U-Net, although displaying some improvements in the F1 score and precision, performed substantially worse in terms of accuracy. This final score amounted to only 77.59%. The lower final accuracy and the higher variance in validation loss indicate that U-Net may not be yet adequately optimized for this particular application: further tuning of its architecture and training regimen is necessary.

The findings from this research depict that SegNet and Vi-U-Net are highly promising in landslide detection, with SegNet being the model of choice based on its superior accuracy. It could be a good foundation for effective early-warning systems and effective tools of geospatial monitoring in landslide regions. Although U-Net is promising in some aspects of the performance metrics, the overall limitations indicate that further and detailed research and optimization might be required. Future work should hone U-Net or explore hybrid models that leverage strengths from SegNet and Vi-U-Net to further improve general detection accuracy and reliability.

This research adds value to knowledge in deep learning in remote sensing and disaster management. This paper clearly offers a pathway for future research work that seeks to enhance landslide detection systems. These final accuracy rates are; 94.80% for SegNet, 94.00% for Vi-U-Net, and 77.59% for U-Net, which show the considerable potential of advanced deep learning models in environmental monitoring as well as risk mitigation.

This study validates the efficacy of deep learning for landslide classification and provides a basis for future developments in predictive modeling. It can be applied in real-time monitoring systems, hence playing a crucial part in reducing disaster risk, lowering response times, thus sometimes reducing landslide impacts on vulnerable communities. It should be further developed to make it scalable and applicable in diverse regions and datasets.

6. References

[1] Sreelakshmi, S., Vinod Chandra, and Shaji E. 2022. "Landslide Identification Using Machine Learning Techniques: Review, Motivation, and Future Prospects."EarthScienceInformatics. https://doi.org/10.1007/s12145-022-00889-2.

[2] Acharya, Vasundhara, Anindita Ghosh, Inwon Kang, Thilanka Munasinghe, and Binita KC. "Landslide Likelihood Prediction Using Machine Learning Algorithms." Rensselaer Polytechnic Institute and NASA Goddard Space Flight Center.

[3] Azarafza, Mohammad, Mehdi Azarafza, Haluk Akgün, Peter M. Atkinson, and Reza Derakhshani. "Deep Learning-Based Landslide Susceptibility Mapping."ScientificReports. https://www.nature.com/scientificreports.

[4] Yang, Zhi-qiang, Wen-wen Qi, Chong Xu, and Xiao-yi Shao. "Exploring Deep Learning for Landslide Mapping: A Comprehensive Review." China Geology. https://www.sciencedirect.com/journal/china-geology.

[5] Zhang, Qi, and Teng Wang. "Deep Learning for Exploring Landslides with Remote Sensing and Geo-Environmental Data: Frameworks, Progress, Challenges, and Opportunities."

[6] Ghorbanzadeh, Omid, Hejar Shahabi, Alessandro Crivellari, Saeid Homayouni, Thomas Blaschke, and Pedram Ghamisi. "Landslide Detection Using Deep Learning and Object-Based Image Analysis."

[7] Omkar, Rukmini Nazre, Soham Naigaonkar, Suraj Sawant, and Himadri Vaidya. "Comparative Analysis of CNN-Based Deep Learning Models for Landslide Detection." COEP Technological University, Pune, India, and Graphic Era Hill University, Dehradun, India.

[8] Devi, Naveen, Dr. Nirmala M., Dummala Roopesh, J. Kiran Reddy, and P. Karthik Raju. 2022. "Landslide Detection Using Machine Learning Algorithms." Journal of Algebraic Statistics 13, no. 3: 2822-2828. https://publishoa.com.

[9] Kainthura, Poonam and Neelam Sharma. “Hybrid Machine Learning Approach for Landslide Prediction”, Uttarakhand, India. ScientificReports. https://www.nature.com/scientificreports.

[10] Tehrani, Faraz S., Michele Calvello, Zhongqiang Liu, Limin Zhang, and Suzanne Lacasse. “Machine Learning and Landslide Studies: Recent Advances and Applications.” ScientificReports. https://www.nature.com/scientificreports.

[11] Youssef, K., K. Shao, S. Moon, and L.-S. Bouchard. “Landslide Susceptibility Modeling by Interpretable Neural Network”.

[12] Tehrani, F. S., Santinelli, G., & Herrera Herrera, M. (2021). MultiRegional landslide detection using combined unsupervised and supervised machine learning. Geomatics, Natural Hazards and Risk, 12(1), 1015-1038.

[13] Wang, H., Zhang, L., Yin, K., Luo, H., & Li, J. (2021). Landslide identification using machine learning. Geoscience Frontiers, 12(1), 351- 364.

[14] Ghorbanzadeh, O., Blaschke, T., Gholamnia, K., Meena, S. R., Tiede, D., & Aryal, J. (2019). Evaluation of different machine learning methods and deep-learning convolutional neural networks for landslide detection. Remote Sensing, 11(2), 196.

[15] Yang, S., Wang, Y., Wang, P., Mu, J., Jiao, S., Zhao, X., ... & Zhu, Y. (2022). Automatic identification of landslides based on deep learning. Applied Sciences, 12(16), 8153.

[16] Kirillov, A., He, K., Girshick, R., & Dollar, P. (2017). A unified ´ architecture for instance and semantic segmentation. In CVPR.

[17] Ghorbanzadeh, O., Xu, Y., Ghamisi, P., Kopp, M., & Kreil, D. (2022). Landslide4sense: Reference benchmark data and deep learning models for landslide detection. arXiv preprint arXiv:2206.00515.

[18] Ghorbanzadeh, O., Blaschke, T., Gholamnia, K., Meena, S. R., Tiede, D., & Aryal, J. (2019). Evaluation of different machine learning methods and deep-learning convolutional neural networks for landslide detection. Remote Sensing, 11(2), 196.

[19] Ji, S.; Yu, D.; Shen, C.; Li, W.; Xu, Q. Landslide detection from an open satellite imagery and digital elevation model dataset using attention boosted convolutional neural networks. Landslides 2020, 17, 1337–1352. https://doi.org/10.1007/s10346-020-01353-2

[20] Ding, A., Zhang, Q., Zhou, X., & Dai, B. (2016, November). Automatic recognition of landslide based on CNN and texture change detection. In 2016 31st Youth Academic Annual Conference of Chinese Association of Automation (YAC) (pp. 444-448). IEEE.

[21] Li, Y., Wang, P., Feng, Q. et al. Landslide detection based on shipborne images and deep learning models: a case study in the Three Gorges Reservoir Area in China. Landslides 20, 547–558 (2023). https://doi.org/10.1007/s10346-022-01997-2

[22] Ronneberger, Olaf & Fischer, Philipp & Brox, Thomas. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. LNCS. 9351. 234-241. 10.1007/978-3-319-24574-4 28.

[23] Zhao, H.; Shi, J.; Qi, X.; Wang, X.; Jia, J. Pyramid scene parsing network. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Honolulu, HI, USA, 21–16 July 2017; pp. 2881–2890 [21] Chandra, N. & Vaidya, H. Building detection methods from remotely sensed images. Current Science. pp. 1252-1267 (2022)

[24] Yun, L., Zhang, X., Zheng, Y., Wang, D., & Hua, L. (2023). Enhance the accuracy of landslide detection in UAV images using an improved Mask R-CNN Model: a case study of Sanming, China. Sensors, 23(9), 4287.

[25] Lin, T. Y., Dollar, P., Girshick, R., He, K., Hariharan, B., & Belongie, S. ´ (2017). Feature pyramid networks for object detection. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 2117-2125).

[26] Kubo, S., Yamane, T., & Chun, P. J. (2022). Study on Accuracy Improvement of Slope

Failure Region Detection Using Mask R-CNN with Augmentation Method. Sensors, 22(17), 6412.

[27] Chandra, N., Ghosh, J. & Sharma, A. A cognitive framework for road detection from high-resolution satellite images. Geocarto International. 34, 909-924 (2019)

[28] Chaurasia, A., & Culurciello, E. (2017, December). Linknet: Exploiting encoder representations for efficient semantic segmentation. In 2017 IEEE visual communications and image processing (VCIP) (pp. 1-4). IEEE.

[29] Shiori, Kubo., Tatsuro, Yamane., Pang-jo, Chun. (2022). Study on Accuracy Improvement of Slope Failure Region Detection Using Mask R-CNN with Augmentation Method. Sensors, doi: 10.3390/s22176412

7. Appendix

Code Link:

https://github.com/Sudhanva7409/DL-Landslide-Detection